Teacher Implementation & Realization of Place in Rural Science Education

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Role of Place within Instructional Context

Resource Levels of *Place within Instructional Context*

Macro
Group

1. Local Generational Knowledge
2. City Parks
3. Community Partnerships

Meso
Group

1. School Resources
2. Library Resources
3. Teacher Collaborations

Micro
Individual

1. Classroom
2. Student Knowledge
3. Student Interest

Teachers experience place at *individual* or *group* levels. Perception of place as a resources are perceived through lens of instructional context
Scannell & Gifford 2010
Bennion et, al., 2022

Rationale

Study Purpose

- Explore how rural teachers drew upon place experiences to define resources and implement a PBE unit

Research Question

- How do rural middle school teachers define their experiences within place?
- What affordances or constraints do they identify within their learning environments?
- **How do these affordances or constraints impact PBE implementation?**

Study Participants

	6-8th STEM	6-8th Science	9th Science	9th Environment
n	5	8	1	1
Teacher Names	Asher Oliver Paul Theodora Wilma	Alice Charlie Chrissy Doreen Fran George Marjorie Sam	Sanders	Shari

- **Exploratory qualitative study**
- **Middle-school place-based unit focus on energy systems**

Study Context

Table 1. *Unit Outline*

Module Title	**Description**
Introducing Building Energy Systems	Building walk to consider energy entry through external sources, energy flow, and energy outputs. Discussions include renewable and non-renewable energy resources, costs, and inequities.
Lighting & Heating	Investigating light and heat energy in different building places, such as measuring light and temperature differences in and across the building to consider energy efficiency.
Engineering a Building	Using building energy knowledge to design and build a one room energy efficient schoolhouse.

Data analysis

		MODES OF INTEGRATION FOR PLACE-BASED EDUCATION													
		Classroom Level (micro)			School Level (meso)		Society- Community Level (macro)								
							connection to students' houses	connection to teachers' houses	connection to surrounding and broader community						
Participants	pseudonym	using classroom windows to teach energy (light and thermal)	pointing electrical pieces: outlet boxes, breaker boxes, wires	using lux meter to explore light energy in classroom	Surface level- Standard school building walk	Deep school building walks			community walk (targeting solar panels, wires, roofs etc.)	discussing about local power plants and grid	discussing about local solar farms	discussing the local politics (county level) and projects like new windmill and solar farms	guest speaker from local community	posting activities or projects on Facebook to share with families and community	TOTAL
1	Alice	1	1			1	1				1		1		**6**
2	Shari	1	1		1		1				1	1			**6**
3	Marjorie	1	1	1		1	1								**5**
4	Paula	1		1		1			1					1	**5**
5	Sam	1		1		1	1						1		**5**
9	Wilma		1			1				1	1	1			**5**
6	Theodora	1	1	1		1									**4**
7	Sanders	1	1	1		1									**4**
8	George	1		1		1				1					**4**
10	Doreen				1				1				1	1	**4**
11	Chrissy			1		1		1							**3**
12	Asher		1		1										**2**
13	Oliver			1	1										**2**
14	Charlie				1										**1**
15	Harriet				1										**1**
16	Paul				1										**1**
17	Fran												1		**1**

Data analysis

MODES OF INTEGRATION FOR PLACE-BASED EDUCATION	**Classroom Level (micro)**		using classroom windows to teach energy (light and thermal)
			joining electrical pieces: outlet boxes, breaker boxes, wires
			using lux meter to explore light energy in classroom
	School Level (meso)		Surface level - Standard school building walk
			Deep school building walks
	Society-Community Level (macro) - Connection to students' houses	**connection to students' houses**	
		connection to teachers' houses	
		connection to surrounding and broader community	community walk (targeting solar panels, wires, roofs etc.)
			discussing about local power plants and grid
			discussing about local solar farms
			discussing the local politics (county level) and projects like new windmill and solar farms
			guest speaker from local community
			posting activities or projects on Facebook to share with families and community

Coding Example

Established Context TCSR

Alter in network

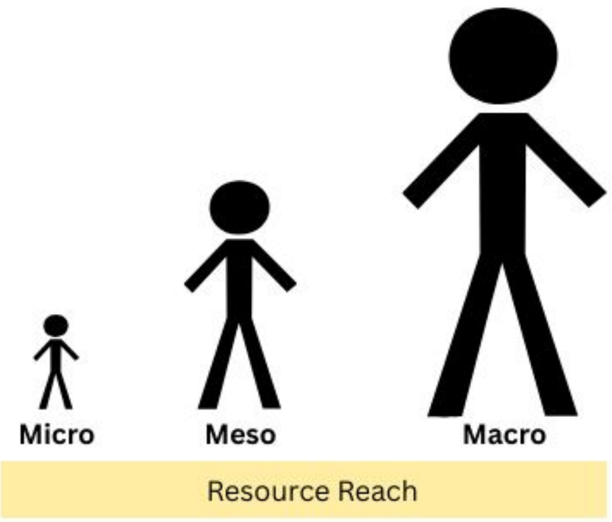
"I took the liberty to review our district's science resource, Inspire Science and I concluded that if we use this resource in all its capacity with allow some areas throughout the unit for students to be true scientists. Individuals who create hypotheses based on their background knowledge. Even with only 20 minutes of my day dedicated to science, if I have more autonomy with how I implement science and have more time, it can be attainable."

Projective Agency

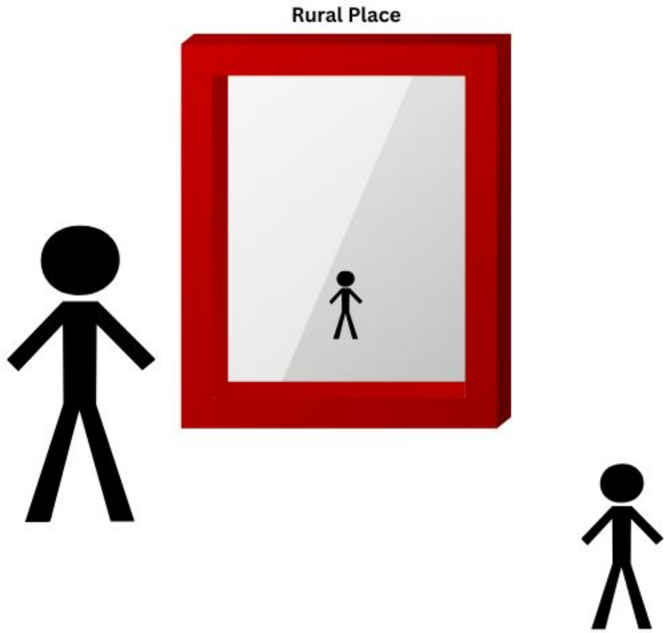
Engagement with a context

Findings

Teachers perceived varied levels of
resource reach



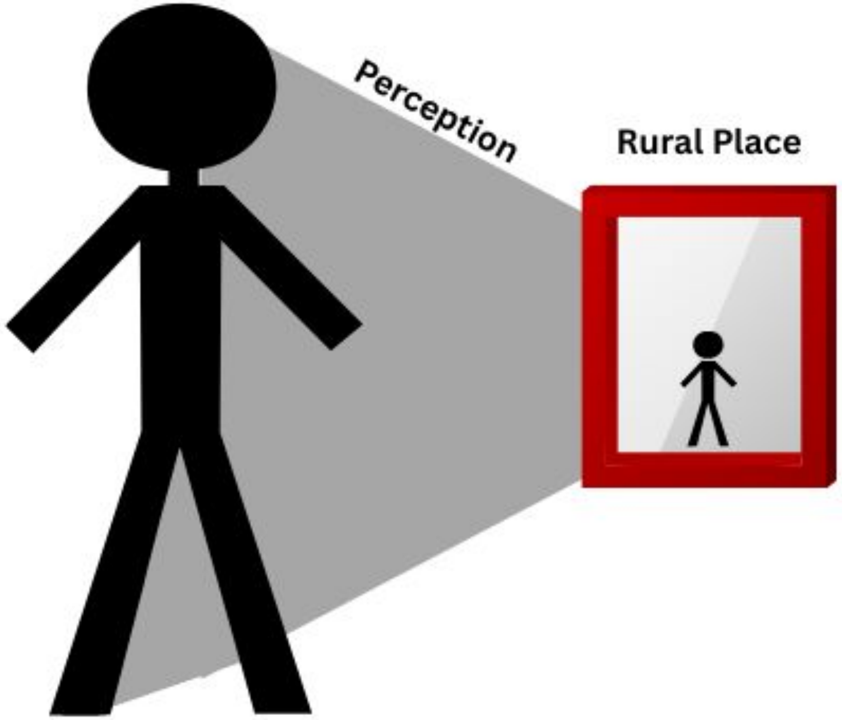
Teachers perceived varied levels of
place integration



Teacher perception of place and resources impacted ***PBE implementation***

Place Disconnected with Constraining Resources

- “We have to use textbooks because PBE doesn’t meet state and national science standards” - Fran & Harriet
- “I’m not even sure the principal noticed what we do.” - Charlie
- “[Students] were all farm kids” and “could care less where energy comes from.” - Paul



Rural Place

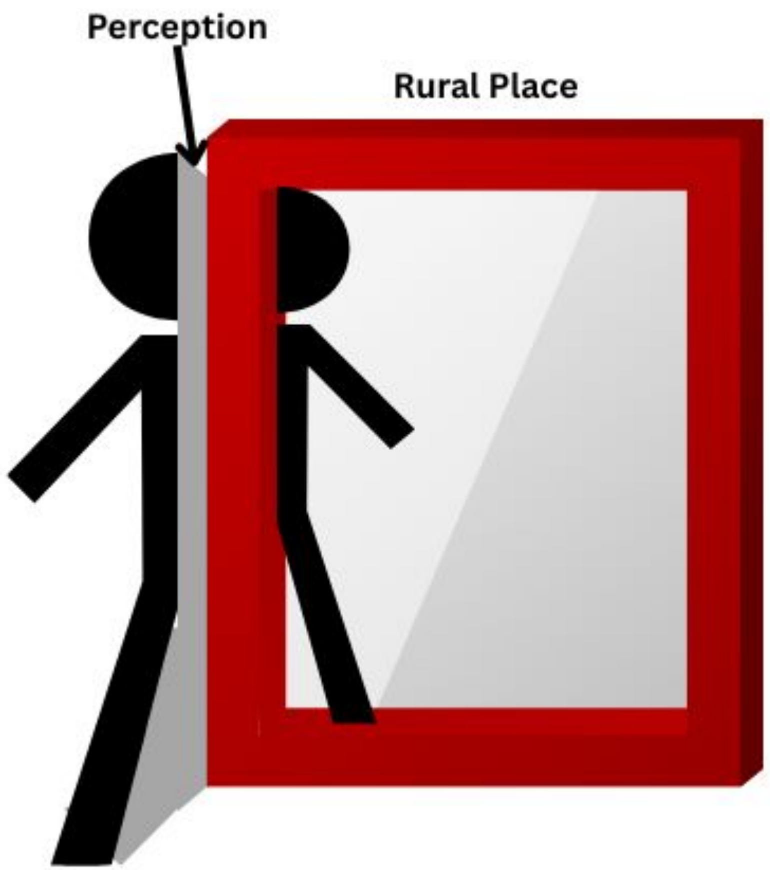
Perception

Place Connected and Disconnected with Shifting Resources

During the unit, seven teachers shifted their perceptions of resource constraints to affordances

Wilma, before building walk:
- “I’m a paper and pencil teacher”.
- “Let’s see how it goes…”

Wilma, after building walk:
- Students asked “good” questions
- Invited janitor to lead a second tour



Place Connected with Resource Affordances

Teachers defined their experiences as embedded within their rural communities

- "Well, since we're rural" - Alice
- "When I talked about breaker boxes… We would go find a breaker box." - Marjorie & Theodora
- Toured local power plants, discussed community issues (e.g., using local land for solar farms)

Rural Place

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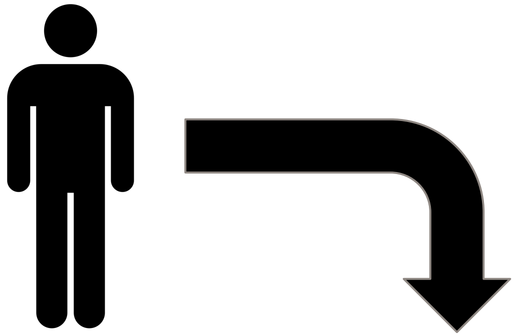

Constraints → Disconnection
Micro
(-) Students were not interested
Meso
(-) Lack of administrator support
Macro
(-) Rural values are “different” (-) Adherence to textbooks (standards)

Rural Place

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Constraints + Experience → Affordances

Micro
(+) Drop-in visits and feedback from admin
Meso
(+) Ample admin support (-) Managing student behavior in hallways (-) Reactions by other teachers
Macro
(-) Lack of resources

"Good" student questions

Inviting janitor to lead building tour

Expanding PBE implementation into the school building

Place Connected with Resource Affordances

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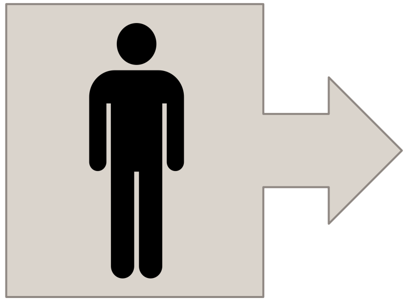

Connection to place transforms the potential for PBE

"Since we're rural…" → Affordances

Micro
(+) Drop-in visits and feedback from admin
Meso
(+) Ample admin support (+) Used school as vehicle, not venue
Macro
(+) Invited community members (+) Toured local power plants and solar farms (+) Invited parents to discuss energy use (+) Community issues (solar farms, etc.)

"… we're rural."

Our building is a teaching tool

Being rural provides unique opportunities and connections

Summary of Findings

- The findings reflect how…
  - Working in a rural school does not mean a teacher will be able to (or desire) to situate their teaching as place-based
  - Teachers' connections to their school and community can transform how they perceive their contexts

Discussion and Implications

- Effective PBE in rural schools involves more than teaching in a rural school; it requires a perceived alignment with place
- Perceptions develop through experiences with place which are interpreted through resource affordances and constraints
- Literature on place attachment suggests that person-place bonds can be fostered through both socially and physically based attachments (Scannell & Gifford, 2010).
- **PBE curricular "prologues" which include place-oriented resource exploration may provide a foundation for place connection, a necessity for PBE implementation**

Thank you!

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References

Azano, A. (2011). The possibility of place: one teacher's use of place-based instruction for English students in a rural high school. *Journal of Research in Rural Education*, *26*(10), 1-12.

Scannell, L., & Gifford, R. (2010). Defining place attachment: A tripartite organizing framework. *Journal of Environmental Psychology*, *30*(1), 1-10.